



A Marker-Free Motion Capture System Built on Unsynchronized Cameras

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Abstract. Human Motion Capture (MoCap) involves tracking and recording human movements. Image-based MoCap systems have gained significant research interest, particularly for their simplified deployment that eliminates the requirement for reflective markers and wearable sensors. However, the deployment of such systems faces two major challenges: temporal misalignments caused by unsynchronized cameras and the sparsity of detectable features on human bodies. To address these limitations, we propose a marker-free MoCap system built on unsynchronized cameras. For the temporal misalignment issue, we introduce two crucial components: multi-view temporal post-processing and temporal augmentation training, which enable automatic temporal alignment across multiple cameras. To overcome feature sparsity, we define virtual markers as 3D surface points on the human mesh model. Experimental results validate the high performance of our system in global marker localization. The proposed approach presents a scalable and flexible solution for MoCap deployment in unconstrained environments, showing promise for applications in mobile health monitoring and sports training.

Keywords: Motion Capture · Temporal Augmentation · Spatiotemporal Fusion

1 Introduction

Human Motion Capture (MoCap) refers to a technology for tracking and recording human movements. The captured movements can be analyzed or reproduced for applications in domains including healthcare and sports training [1, 2].

Various technologies, such as infrared cameras and Inertial Measurement Units (IMUs), have been deployed into MoCap systems and used in different application scenarios [3]. Employing reflective markers as explicit features, infrared camera-based MoCap systems track the 3D positions of reflective markers to represent human motions, offering high accuracy [4]. Thus, infrared camera-based MoCap systems are usually adopted in indoor applications, such

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as biomechanics and animation production [5]. In contrast, IMU-based systems, which require IMU mounted on different body segments, enable wearable and portable deployments in unconstrained and outdoor settings [6].

Image-based methods provide an alternative approach for human MoCap, offering advantages in deployment simplicity and application scope [7]. By analyzing spatial relationships between body segments, image-based MoCap avoids the need for physical attachments such as reflective markers and wearable sensors, thus avoiding related issues like detachment [8]. Furthermore, image-based MoCap systems show low drift and high adaptability to varying conditions, benefiting from low system noise and well-established image processing methods [9]. These advantages make image-based MoCap systems adaptable to broad application scenarios, from indoor to outdoor spaces [10]. Recent studies have addressed the limitations of monocular cameras, including occlusion problems and insufficient depth constraints, through multiple camera deployment [11]. However, the approach based on multiple cameras increases deployment complexity [12].

Temporal misalignment between cameras, termed *unsynchronized cameras*, affects the deployment of image-based MoCap systems [13]. While hardware-level synchronization between multiple cameras is essential, it typically requires close-range wired connections, thus limiting the operational range of multi-camera image-based MoCap systems [14]. For spatially distant cameras that primarily rely on network connections, achieving precise hardware synchronization becomes particularly challenging [15]. In the absence of hardware synchronization capabilities, researchers typically resort to network-based synchronization or post-processing approaches, such as synchronization actions [16]. Network synchronization is susceptible to network latency, while post-processing methods increase manual workload and prevent real-time MoCap [17]. This synchronization issue results in inconsistent observations of the same motion event across different cameras, compromising the performance of multi-camera image-based MoCap systems [18].

The marker-free characteristic also affects the deployment of image-based MoCap systems [19]. Instead of physical markers, image-based methods utilize either 2D texture features, such as SIFT, or 3D geometry features, such as lines and planes [20]. However, 3D geometry features are distributed sparsely on the human body [11]. And 2D texture features lack specificity with stationary body segments [21], such as 3D joint positions. To address the challenges, existing image-based MoCap systems extract skeletal coordinates from the images using deep learning-based methods, rather than visual features on surfaces [22]. The implicit mapping from surface to skeletal coordinates may increase training complexity and the instability of multi-view fusion, potentially affecting system deployment [23].

In this paper, to address the deployment complexity caused by unsynchronized cameras and the marker-free characteristic of image-based MoCap systems, we propose a MoCap system (1) aligning the timestamps between cameras automatically, and (2) optimizing surface-level features directly. By incorporating temporal augmentation during training and a multi-view temporal post-

processing strategy, we enable robust fusion across unsynchronized views. To mitigate feature sparsity, we introduce virtual markers which are 3D surface points located near the target physical markers. They can serve as more consistent and informative spatial cues.

The contributions of this paper are listed as:

- (1) We propose a MoCap system that automatically aligns the timestamps between unsynchronized cameras. With temporal augmentation training and multi-view temporal post-processing, the proposed system can deal with the temporal misalignment.
- (2) We employ human virtual marker extraction to enhance feature information, thereby improving the accuracy of the proposed marker-free MoCap system.
- (3) We conduct comprehensive experiments to validate the performance of the proposed marker-free MoCap system. Various human motion types, including walking, running, and in-place motions, are included in our experiments.

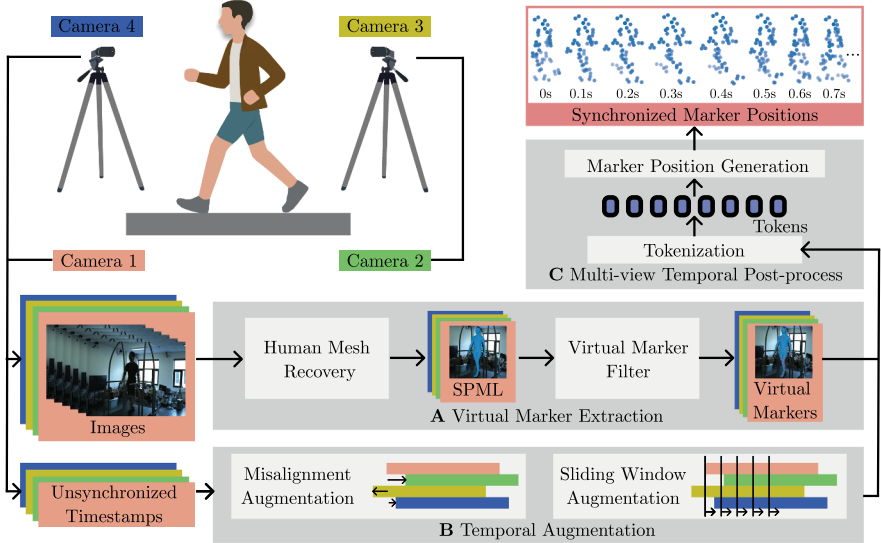


Fig. 1. Pipeline of the proposed MoCap system. **A** shows the 3D human mesh for each frame is recovered, from which virtual markers are filtered by a manually designed mapping to enhance feature information. **B** illustrates that we incorporate temporal data augmentation during training to enhance the model’s temporal alignment capability. **C** presents our transformer-based post-process model that fuses multi-view and temporal camera information within a temporal window.

2 System Definition

Our MoCap system takes multiview images with unsynchronized timestamps as inputs. The system then obtains synchronized 3D coordinates of predefined human body landmarks from these images.

Given N unsynchronized cameras $C = \{c^1, c^2, \dots, c^n\}$, where c^1 is designated as the main camera, each camera c^i has intrinsic parameters K^i , sampling characteristics f^i and σ^i . $K^i \in \mathbb{R}^{3 \times 3}$ is the intrinsic matrix, including focal length and optical center. f^i and σ^i are the frame rate and capture time variance. Each camera c^i is also characterized by its extrinsic parameters (R^i, T^i) , where $R^i \in \mathbb{R}^{3 \times 3}$ and $T^i \in \mathbb{R}^3$ define the transformation from the coordinate system of camera c^i to that of the main camera. Let $x^i \in \mathbb{R}^3$ be a 3D point in the camera c^i coordinate system and $\mathbf{x} \in \mathbb{R}^3$ be its corresponding coordinate in the world coordinate system. The transformation between these coordinates is given by:

$$\mathbf{x} = R \cdot (R^i x^i + T^i) + T, \quad (1)$$

where $R \in \mathbb{R}^{3 \times 3}$ and $T \in \mathbb{R}^3$ define the transformation from the main camera coordinate to the world coordinate, which is unknown.

Each camera c^i captures an image sequence $\{I_j^i\}$ at its own timestamps $\{t_j^i\}$. Timestamps are not synchronized across different cameras. Each camera's timestamp has an unknown fixed bias B^i from the global timestamp.

$$\tau_j^i = t_j^i + B^i, \quad (2)$$

where $\tau_j^i \in \mathbb{R}$ is the global timestamp for image I_j^i . We can assume that $|B^i| \leq \epsilon$, $B^1 = 0$, and $t_{j+1}^i - t_j^i \sim \mathcal{N}(1/f^i, (\sigma^i)^2)$, where ϵ is synchronization time for system.

Consider we have a defined marker set M , an expected total MoCap time $\mathbf{t}_c$, and required output frequency $\mathbf{f}$. Each marker is attached to the human body. Our goal is fusing all unsynchronized and multi-view information, and calculating the synchronized global position of the defined marker set $Y = \{y_i = [\mathbf{x}_0^{m,i}, \mathbf{x}_1^{m,i}, \mathbf{x}_2^{m,i}]\}_{m \in M, i \in [0, \mathbf{t}_c \times \mathbf{f}]}$, where $\mathbf{x}^{m,i} \in \mathbb{R}^3$ represents the global position of marker m in global time $i/\mathbf{f}$.

We can cut the image sequence into time window δ to divide the problem into smaller problems,

$$W_d = \{(I_j^i, t_j^i) | d\delta \leq t_j^i \leq (d+1)\delta\}. \quad (3)$$

For each window W_d , we calculate the results $\hat{Y}_d = \{\hat{y}_i\}$. The overall result is obtained by aggregating all window outputs: $\hat{Y} = \bigcup_d \hat{Y}_d$, where $\hat{Y}$ represents the final set of predictions.

3 Methodology

As shown in Fig. 1, we propose a MoCap system that features three key components: temporal augmentation for improved tolerance to timing misalignments, virtual markers near the body surface for enhanced spatial features, and a transformer-based fusion model for processing unsynchronized multiview data.

3.1 Virtual Marker Extraction

To obtain richer feature information and achieve better MoCap performance in marker-free settings, we extract and filter the coordinates of virtual markers located near the target of the human mesh and use them as input for the next stage. The 3D human mesh for each frame is estimated using advanced human mesh recovery methods [24], which are represented by f_{HMR} and can be defined as

$$V_j^i, J_j^i, \beta_j^i = f_{\text{HMR}}(I_j^i, K^i). \quad (4)$$

In Eq. (4), $V_j^i = \{p_{j,k}^i \in \mathbb{R}^3\}$ represents the estimated human mesh vertices, each $p_{j,k}^i$ represents the position of a vertex in coordinate in camera c^i . $J_j^i = \{r_{j,k}^i \in \mathbb{R}^3\}$ represents the estimated human joints rotation, each $r_{j,k}^i$ represents a rotation vector of a joint in the coordinate of camera c^i . The shape parameters β_j^i control the body's physique, muscle structure, and other characteristics, and are represented as a ten-dimensional vector.

A fixed mapping $\mathcal{K}$ is manually defined between virtual markers and mesh vertices on the Sparse Principal Manifold Learning (SPML) human model, based on empirical selection according to their spatial proximity to the target physical markers. These selected vertices, virtual markers, serve as key visual features for subsequent processing.

3.2 Multi-view Temporal Post-process

We propose a transformer-based multi-view human motion temporal post-process model [25]. After receiving feature information from each camera, the system first tokenizes the data and then passes it through a transformer-based network to obtain synchronized coordinates of target marker points.

Each window W_d is tokenized into a discrete representation X_d . Each token encodes a single observation from one of the n unsynchronized cameras, and is constructed to include all relevant information projected into the main camera coordinate system. Formally, each token $\mathbb{T}_j^i \in X_d$ is a vector comprising the following components:

$$\mathbb{T}_j^i = \left[\text{OneHot}(i), t_j^i/\delta - d, \beta_j^i, \{R^i \times r_{j,k}^i\}, \{R^i \times p_{j,\mathcal{K}(m)}^i + T^i | m \in M\} \right], \quad (5)$$

where $\{R^i \times r_{j,k}^i\}$ represents joint position features, $\{R^i \times p_{j,\mathcal{K}(m)}^i + T^i | m \in M\}$ denotes virtual marker features, and all tokens are sorted and embedded temporally using their normalized timestamps $t_j^i/\delta - d$. We further apply sinusoidal positional encoding (using cosine-based functions) across the entire token sequence to capture global temporal dependencies.

The transformer model with parameters w processes these sequential representations, capturing spatiotemporal dependencies and generating tokens on marker position $\hat{Y}_d$:

$$\hat{Y}_d = \text{Transformer}(\{\mathbb{T}_j^i\}, w). \quad (6)$$

Detailed model specification can be found in Sect. 4.

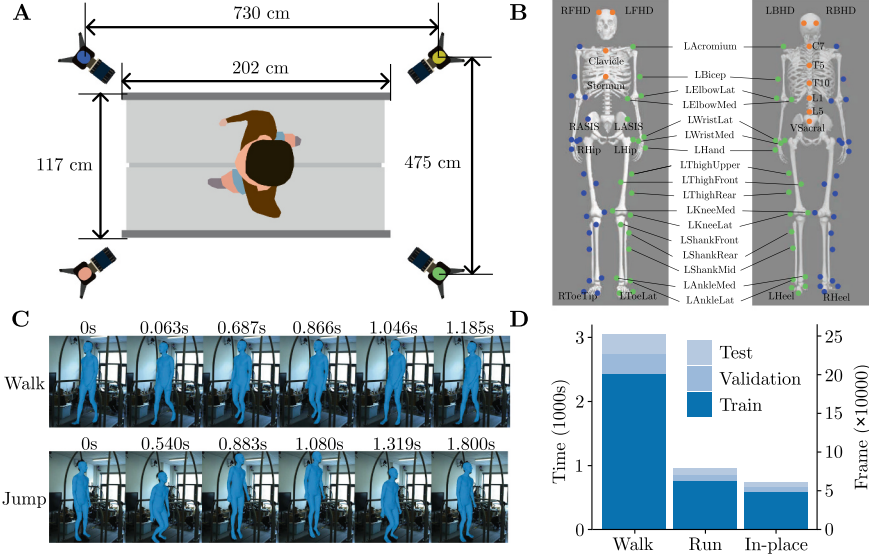


Fig. 2. Experiment Setup. **A** shows schematic diagram of the experimental setup. **B** presents marker set M used for implementation and analysis. **C** illustrates example motions, including walking and jumping. **D** provides dataset distribution categorized by walking, running, and in-place actions, along with the division into train, validation, and test sets.

3.3 Temporal Data Augmentation

To enhance the model’s robustness to temporal synchrony, we introduce a temporal data augmentation strategy based on synthetic timestamp bias. High-precision optical MoCap is used as Ground-Truth (GT) marker trajectories, represented by $Y = \{y_i\}$.

Misalignment Augmentation: To simulate real-world unsynchronized capture scenarios, we introduce synthetic temporal offsets into each camera stream during training. For each camera c^i , we randomly draw a temporal bias $\tilde{B}^i \sim \mathcal{U}(-\epsilon, \epsilon)$ for each training instance and adjust the original timestamps accordingly.

$$\tilde{t}_j^i = \tau_j^i + \tilde{B}^i. \quad (7)$$

Sliding Window Augmentation: To further enrich the training data and increase temporal coverage, we randomly select multiple possible starting times $s \sim \mathcal{U}(0, \delta)$ indices within each action sequence. This strategy ensures that the model observes varied temporal contexts, motions at different phases, and diverse cross-camera alignments. The training window is constructed as

$$\tilde{W}_d = \{(I_j^i, \tilde{t}_j^i) | d\delta \leq \tilde{t}_j^i - s \leq (d+1)\delta\}. \quad (8)$$

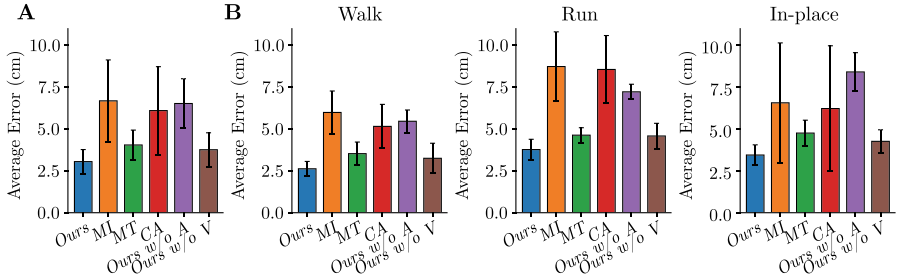


Fig. 3. Accuracy. **A** shows average error comparison across all methods, with error bars indicating the standard deviation. Our method shows the lowest error overall. **B** shows average error under different motion types.

For all actions in action space, we use this data augmentation method and get a dataset $\{\tilde{W}_d, Y_d\}$. The transformer-based post-process network is trained on this dataset using a Mean Squared Error (MSE) loss function between the predicted marker positions and the ground truth from the MoCap system at global timestamps.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|Y_d|} \sum_{y_i \in Y_d} \|\hat{y}_i - y_i\|_2^2. \quad (9)$$

4 Experiment Setup

System Implementation: Our MoCap system is implemented in an indoor environment using a four-camera setup ($n = 4$) as illustrated in Fig. 2A. Intrinsic (K^i) and extrinsic (R^i , T^i) camera parameters are calibrated using Zhang’s method [26]. To enhance temporal asynchronization, cameras operate at different frame rates: $f^1 \approx f^2 \approx 30$ Hz and $f^3 \approx f^4 \approx 10$ Hz, with timing uncertainty $\sigma^i \approx 0.2$. The system employs a sliding window ($\epsilon = 0.2s$, $\delta = 2s$) and outputs at $f = 50$ Hz. GT temporal biases are established through hand-waving calibration. The marker set comprises $|M| = 56$ whole-body landmarks, visualized in Fig. 2B using OpenSim [27].

Data Collection: To train and validate our MoCap system, we build a dataset containing a diverse set of human motions. Motions can be categorized into three primary types: walking, running, and in-place actions. For the walking and running categories, participants performed motions at five distinct speeds. The in-place action category includes repetitive movements such as squatting, jumping, stretching in place, recorded while facing different orientations. We temporally segment the collected motion data into training, validation, and test sets using an 8:1:1 ratio. Detailed dataset information is shown in Fig. 2C and Fig. 2D. We use Vicon MoCap system (Vicon Valkyrie V16, 12 infrared cameras) to collect GT 3D global marker trajectories.

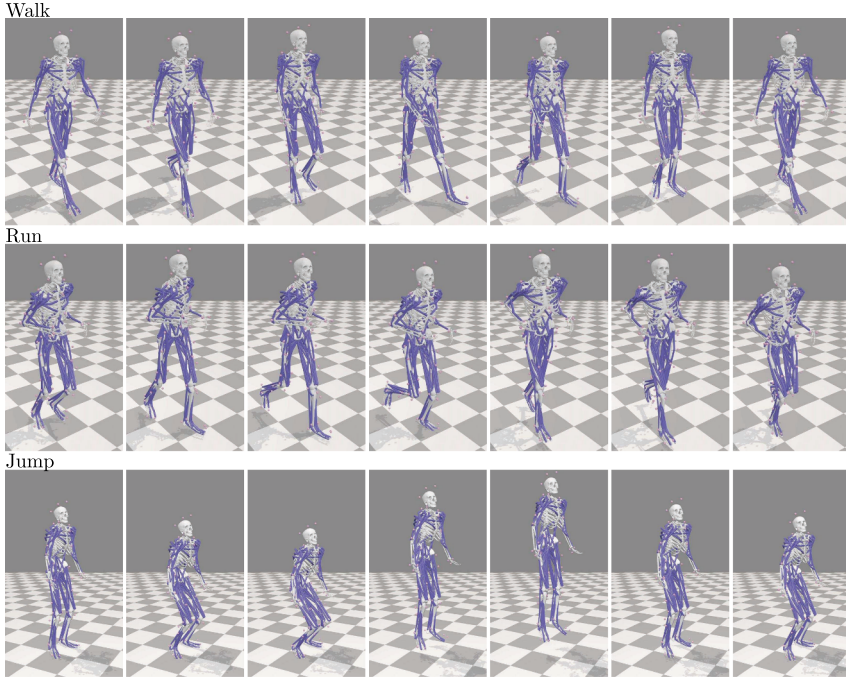


Fig. 4. Skeleton showcases. Utilizing the marker positions estimated by our MoCap system, we reconstruct skeletons with OpenSim. These showcases demonstrate that the output marker positions can be used to restore various human motions, including walking, running, and jumping.

Experimental Protocol: We evaluate system performance using the Average Error metric, defined as the average distance between predicted and GT global marker positions. This metric is crucial for both system accuracy assessment and downstream applications such as dynamic analysis and biomechanical computations.

We benchmark our MoCap approach (*Ours*) against several baseline configurations. The main camera interpolation baseline (*MI*) processes observations exclusively from the main camera, while the main camera training variant (*MT*) limits both training and inference to main camera data. The cameras average baseline (*CA*) implements naive multi-view fusion across all streams. To assess individual components, we evaluate two ablation variants: one without data augmentation (*Ours w/o A*) and another without virtual markers (*Ours w/o V*, using only joint position features).

The post-processing transformer architecture employs a 4-layer network with 8 attention heads and a hidden dimension of 512. The model is trained for 1,000 epochs using a batch size of 256, with a dropout rate of 0.1 for regularization. Optimization is performed using Adam with an initial learning rate of

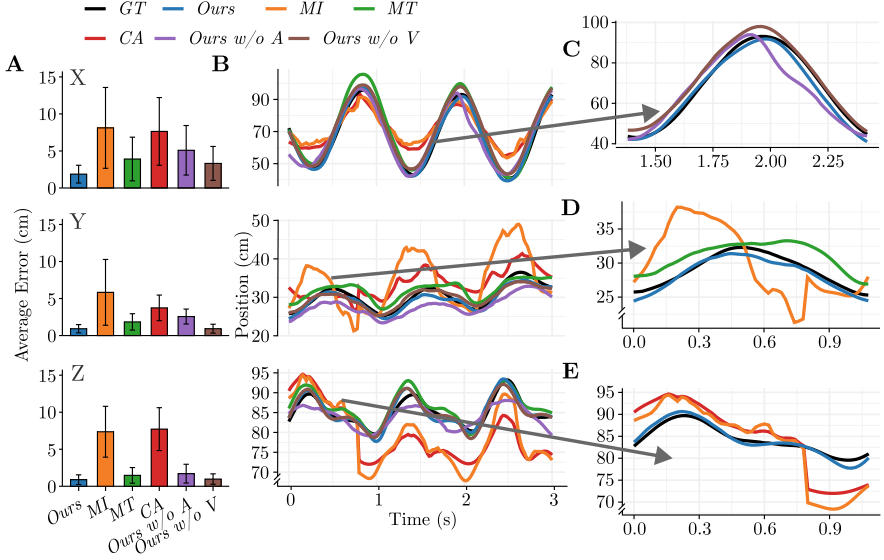


Fig. 5. Marker trajectory showcases. **A** and **B** show the tracking results of various methods for the **LHand** virtual marker during a time period in the test set under the Walk (1.2m/s). Specifically, **A** presents the average error of each method along different axes during this period, where *Ours* consistently achieves the lowest error. **B** illustrates a detailed view of its trajectory. **C** shows *Ours w/o V* and *Ours w/o A* results in increased spatial errors and noticeable temporal lag, respectively. **D** shows *MT* improves temporal consistency over *MI* via post-processing, but its accuracy remains inferior. **E** shows *MI* and *CA* suffer from severe temporal jitter and spatial deviations.

5×10^{-5} and a step decay scheduler ($\gamma = 0.5$) applied at 200-epoch intervals. All experiments are executed on a single NVIDIA RTX 3090 GPU.

5 Experiment Results

As shown in Fig. 3A, our method achieves the lowest overall error of 3.05 cm, compared to 6.68 cm for *MI*, 4.04 cm for *MT*, and 6.10 cm for *CA*. Removing either data augmentation (*Ours w/o A*) or virtual marker features (*Ours w/o V*) led to higher errors of 6.52 cm and 3.76 cm. Figure 3B directly compares the performance of *Ours* and its ablated variants across walking, running, and in-place motions. *Ours* yields a mean error below 4 cm, and substantially outperforming *MI*, *MT*, and *CA*. For Walk, our method achieves 2.62 cm, versus 5.99 cm (*MI*), 5.46 cm (*Ours w/o A*), and 5.16 cm (*CA*). For Run, the error increases for all models, but *Ours* remains lowest at 3.77 cm, while *MI* and *CA* exceed 8.5 cm. For in-place motion, our model records 3.44 cm, with the other variants ranging from 4.26 cm (*Ours w/o V*) to 8.41 cm (*Ours w/o A*). Our model achieves consistently lower errors than all baselines across the three motion categories.

Please refer to Fig. 4 for skeleton motion reconstructed from *Ours*. For marker trajectory showcase, as shown in Fig. 5A and B, *Ours* achieves the most stable and accurate tracking of the LHand marker in a walking motion (1.2 m/s), outperforming prior baselines.

6 Discussion

This paper introduces a marker-free motion capture system designed for unsynchronized multi-camera setups. The proposed approach incorporates two key techniques: temporal augmentation and virtual marker extraction, enabling automatic camera synchronization. Comprehensive ablation studies and baseline comparisons validate the effectiveness of these components. Removing the virtual marker features (*Ours w/o V*) increases spatial prediction errors (average 0.71 cm), and omitting temporal augmentation (*Ours w/o A*) introduces temporal drift and phase gap (see Fig. 5C). These results confirm the necessity of both virtual marker extraction and temporal augmentation for achieving robust spatial and temporal performance. Although *MT* applies post-processing to enhance temporal consistency compared with *MI*, its accuracy is still limited by the absence of multi-view camera information, underscoring the importance of multi-view input in MoCap (see Sect. 5 and Fig. 5D). Moreover, both *MI* and *CA* exhibit temporal jitter and large spatial deviations, demonstrating that our model effectively captures and aligns temporal information across unsynchronized cameras (see Fig. 5E). Based on the above analysis, our contributions form a cohesive system that enables accurate, stable 3D human motion reconstruction.

The future research directions can be explored from various aspects. The adaptability of the system to different environments can be enhanced by expanding the training dataset through varying camera models, placement positions, and collecting data in diverse environments [28]. A human body surface model with more adjustable parameters holds the potential for more accurate MoCap results [29,30]. Joint optimization incorporating musculoskeletal models offers potential for effective clinical implementation [31].

In this paper, we present a novel marker-free MoCap system that utilizes multiple unsynchronized cameras, addressing key challenges in fusing multi-view, temporally misaligned images and the sparsity of detectable features on human bodies. The experiment results indicate that our methods have improved both localization accuracy and robustness to temporal misalignment. The proposed MoCap system presents an alternative to existing MoCap systems, making the MoCap system easier to deploy in real-world scenarios.

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